PRACTICAL 16
NFS and CIFS Network Storage

Aim
To access remote NFS/CIFS network storage and configure persistent mounts.

A. NFS Client
Install:
sudo dnf install nfs-utils
List exports from server:
showmount -e SERVER_IP
Mount:
sudo mkdir /mnt/nfs
sudo mount SERVER_IP:/shared /mnt/nfs
Check:
df -h /mnt/nfs

B. NFS in /etc/fstab
Example:
SERVER_IP:/shared /mnt/nfs nfs defaults,_netdev 0 0
Test:
sudo mount -a

C. CIFS
Install:
sudo dnf install cifs-utils
Create mount point:
sudo mkdir /mnt/cifs
Mount:
sudo mount -t cifs //SERVER_IP/share /mnt/cifs -o username=USER
The command will prompt for a password where appropriate.

D. Automount

sudo dnf install autofs
Enable:
sudo systemctl enable --now autofs
Configuration depends on the NFS/CIFS server and mount map being used.

PRACTICAL 17
Shell Scripting

Aim
To create shell scripts for system information, automation and process monitoring.

A. System Information Script
Create:
vi systeminfo.sh
Enter:

```
#!/bin/bash

echo "===== SYSTEM INFORMATION ====="

echo "Hostname:"
hostname

echo "Kernel:"
uname -r

echo "Operating System:"
cat /etc/os-release | grep '^PRETTY_NAME'

echo "Current User:"
whoami

echo "Date:"
date

echo "Uptime:"
uptime

echo "Memory:"
free -h

echo "Disk:"
df -h /
```

Make executable:
chmod +x systeminfo.sh
Run:
./systeminfo.sh

B. Process Monitoring Script
Create:
vi process_monitor.sh
Enter:

```
#!/bin/bash
```

```
echo "===== TOP CPU PROCESSES ====="

ps -eo pid,user,%cpu,%mem,comm --sort=-%cpu | head -n 6

echo
echo "===== TOP MEMORY PROCESSES ====="

ps -eo pid,user,%cpu,%mem,comm --sort=-%mem | head -n 6
```

Run:

```
chmod +x process_monitor.sh
./process_monitor.sh
```

PRACTICAL 18
Scheduling Tasks Using Cron and At

Aim
To schedule recurring and one-time tasks using cron and at.

A. Cron
Check service:
systemctl status crond

Cron format
Minute Hour Day Month Weekday Command
Example:
* * * * * command
means every minute.

Create Cron Job
Edit:
crontab -e
Example:
*/5 * * * * date >> /home/student/cron-output.txt
This writes the date every five minutes.
List cron jobs:
crontab -l

B. at
Install if required:
sudo dnf install at
Enable service:
sudo systemctl enable --now atd
Schedule command:
echo "date >> /home/student/at-output.txt" | at 23:00
List jobs:
atq
Remove a job:
atrm JOB_NUMBER
